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E-481 500 kV COMPRESSED GAS INSULATED BUS

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to site, installation, commissioning and testing of 500 kV Compressed Gas Insulated Bus in accordance with Part 4, Subsection 1.2 Summary of Work.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5 Codes and Standards and the following specific standards:

2.1 Indonesian National Standards

- PUIL : Peraturan Umum Instalasi Listrik-Indonesia (General Regulation of Electric Installation-Indonesia)
SPLN : Standard Perusahaan Umum Listrik Negara.

2.2 International Electrotechnical Commission

- IEC 60060 : High-voltage test techniques
IEC 60071 : Insulation coordination
IEC 60137 : Insulated bushings for alternating voltages above 1000 V
IEC 60376 : Specification and acceptance of new sulphur hexafluoride
IEC 62155 : Hollow pressurized and unpressurized ceramic and glass insulators for use in electrical equipment with rated voltages greater than 1000 V
IEC 62271 : High-voltage switchgear and controlgear

2.3 Institute of Electrical and Electronics Engineers

- IEEE C37.122 : Standard for Gas-Insulated Substations

2.4 American Society for Testing and Materials

- ASTM D2472 : Standard Specification for Sulfur Hexafluoride

3. OPERATING AND DESIGN CONDITIONS

The CGIB shall be designed for outdoor operation in tropical climate along the seacoast. Refer to Part 4, Subsections 1.1 and 2.14, 2.16 for meteorological, seismic and environmental data of the site.

4. DETAILED REQUIREMENT

4.1 General

The CGIB shall be designed to connect the steam turbine generator transformer to the 500 kV conventional substation. The CGIB shall consist of metal enclosed bus with Sulfur Hexafluoride (SF₆) gas as the insulating medium. The CGIB shall be an isolated phase design.

The CGIB equipment will be installed above ground and underground. For the installations above ground, the CGIB shall be designed and installed with an appropriate support system which will not limit access to equipment in the 500 kV substation or to equipment adjacent to the generator transformer. For the underground installations, the CGIB shall be designed with an appropriate support system along the duct.

The CGIB system enclosure shall have enough current carrying capacity and shall be fully bonded to negate short circuit forces and limit exterior magnetic fields. The enclosure shall prevent explosive shattering caused by internal faults. The CGIB shall be furnished with expansion joints, as required, to handle expansion caused by thermal stress due to current or sun heating.

4.2 Ratings

The 500 kV CGIB shall comply with the followings:

- | | | |
|-----|---|--|
| (1) | Rated continuous current..... | 5% above generator transformer rated current |
| (2) | Basic insulation level (BIL) | 1,550 kV |
| (3) | Switching impulse withstand voltage | 1,175 kV |
| (4) | Nominal system voltage | 500 kV |
| (5) | Maximum system voltage | 525 kV |
| (6) | System frequency | 50 Hz |
| (7) | Power frequency withstand voltage..... | 620 kV |
| (8) | Maximum short-circuit current..... | 50 kA |
| (9) | Duration of short-circuit current | 1 second |

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The Contactor shall verify the adequacy of the arresters connected in front of CGIB for protecting the CGIB (with 15% protective margin against the switching

surges and 20% protective margin against the lightning surges).

The CGIB gas monitoring and alarm system shall be designed according to the following criteria :

SF ₆ Gas Density (%)	Action
100	Normal Operation
95 – 90	Initiate first-stage alarm range
85 – 80	Initiate second-stage alarm range
Less than 80	Trip the generator circuit breaker

The CGIB shall be designed with sufficient design margin to maintain full system rating at SF₆ gas density levels from 100 % to 80 %.

The CGIB system controls shall be designed to include an adjustable time-delay (0-10 minutes) relay. The time-delay relay shall be energized by the first stage low gas density alarm. If a second stage low gas density alarm occurs within a period of five (5) minutes, indicating a rapid loss of gas, the time-delay relay should initiate circuit trip, if the pressure is less than 80%.

4.3 Conductors

The CGIB inner conductor shall be of a high-conductivity aluminum alloy or copper.

4.4 Electrical Clearance

Clearance and internal dimensions shall be determined by the Contractor's design and shall be in accordance with relevant industrial standards.

4.5 Field Joints and Elbows

Each field joint shall have conductor plug-in joint that has been completely assembled and tested in the factory before shipment. The conductor plug-in unit shall allow sliding of the contact fingers to take up differential thermal expansion of the conductor relative to the enclosure.

When a welded enclosure joint is employed, it shall consist of an internal backup ring. The backup ring shall contain silicone rubber seals which act as contamination barriers to particles which might otherwise enter the in to the gas bus.

When a bolted construction is used, the bolted joint shall consist of two (2) flanges, one (1) on each section to be bolted together. The flanges shall have double “O-ring” seals (with a leak check port) so that no leak will occur after bolting.

Factory fabricated elbows shall be furnished where required for horizontal or vertical routes.

Joints connecting to equipment such as transformers, surge arresters, and substation equipment shall be bolted. Bolted joints shall be flanged.

4.6 CGIB to Transformer Connections

The Contactor shall furnish all materials required for connection of the CGIB to the generator transformer. Connections at the transformer shall allow at least 25 mm of movement in any direction.

The Contractor shall furnish all corona-free connectors as required to terminate the CGIB to the transformer bushings. The CGIB terminations at the transformer shall be totally enclose the transformer bushing and live parts in the SF₆ gas. Access openings shall be provided in each phase terminal enclosure to permit removal of connectors to isolate the transformer and for insulation tests.

4.7 Grounding

The individual CGIB phase enclosure shall be directly bonded to each other by flexible bonding leads. These bonding leads shall also be the point of connection for the ground grid. Sufficient bonding and grounding points shall be provided in accordance with the Technical Specification E-240 and E-241.

4.8 Supporting Structures

Supporting structures shall be provided for the CGIB. These structures shall be hot-dipp galvanized steel. These structures shall also be painted to match the CGIB final color. Supporting structures shall be fabricated with either tubular or wide flange shapes. Lattice type structures are not acceptable.

4.9 Gas and System Monitoring

The components of the CGIB shall limit gas loss in the event of leakage. Each gas compartment shall not include more than one (1) phase.

Enclosures and gas-tight barriers shall be capable of withstanding 1.5 times max-

imum working pressure. Gas tight barriers shall be clearly marked on the enclosure.

Each gas compartment shall have provisions for filling, sampling, evacuating, filtering the gas. Each gas compartment shall contain of a rupture disk for overpressure relief from extreme pressure. Temperature-compensated gas density relays with two (2) sets of independently variable contacts to provide two (2) levels of alarm shall be provided for each gas compartment in the CGIB to alarm loss of gas. The Contractor shall indicate the settings available for these relays. It shall be possible to remove each gas density relay for checking while the compartment to which it is fitted remains in service.

Each independent gas system shall also include temperature-compensated pressure gauges and moisture level indicators. A non-removable final gas filter shall be provided on all fill ports.

The guaranteed maximum SF₆ gas leakage shall be less than one (1) % per year for each CGIB circuit.

The Contractor shall furnish all SF₆ gas required including an additional 10 % spare SF₆ gas quantity. All gas provided shall be in accordance with ASTM D2472 or equivalent, except as otherwise specified.

All external gas piping shall be stainless steel and shall have protective covers in areas where the piping may be damaged (stepped on, etc.) during normal operation and routine maintenance activities. The gas monitoring piping shall be electrically isolated so that it does not carry electric current during fault conditions.

Each gas filling connection and monitoring equipment cabinet shall be furnished with the followings:

- (1) Temperature compensated pressure gauge per gas compartment.
- (2) Temperature compensated gas density relay per gas compartment.
- (3) Manifold for equalizing pressure.
- (4) Space heaters with thermostat to maintain cabinet temperature above the dew point.
- (5) Terminal blocks.

- (6) Moisture level indicators per gas compartment.
- (7) Visual annunciation system to indicate low gas pressure Stage 1 and Stage 2 for each gas compartment.

4.10 Nameplates

The nameplates for the outdoor 500 kV CGIB shall be stainless steel. The nameplates for equipment/devices inside the monitoring cabinets shall be three-ply laminated plastic.

An overall CGIB nameplate shall be mounted at the monitoring cabinet for each gas compartment with the following information as a minimum :

- (1) Overall circuit gas barrier layout and gas system information (including pressures, alarm settings, etc.)
- (2) Manufacturer's type and serial number
- (3) Year of manufacture
- (4) CGIB ratings including as a minimum
- (5) Temperature ranges
- (6) System frequency
- (7) Maximum continuous voltage and current
- (8) Insulation withstand voltages
- (9) Short circuit currents
- (10) Rated and maximum gas pressure at 30° C
- (11) Recommended minimum gas pressure at 30° C
- (12) Contract order number
- (13) Electrical parameters
- (14) Total weight of gas and bus

Nameplates shall be provided to identify the CGIB electrical phase along the route. These nameplates shall be provided at least at every other support structure along the route. These nameplates shall be attached to the CGIB to allow easy identification from the ground. Lettering on these nameplates shall be at least 75 mm high.

5. PERFORMANCE TESTING AND GUARANTEES

The Contractor shall perform type tests, routine tests and site tests as specified hereinafter. If type test certificate of identical equipment is available than no repeated test is necessary.

The lightning impulse test voltage for the type and routine test shall be 1550 kV.

The partial discharge test shall be performed in accordance with IEC 62271. The guaranteed partial discharge intensity shall be included with the Bid information.

The pressure test shall be performed at 1.5 times maximum working pressure.

High-voltage test shall be performed to check dielectric strength of the completely assemble CGIB. Bidders shall submit a high-voltage test program which shall comply with the following minimum requirements:

- (1) The CGIB shall be erected completely and gas-filled at its rated gas density before the tests.
- (2) The high-voltage tests shall be performed at the end of all other site tests.
- (3) The chosen test procedure shall not jeopardize healthy parts of the CGIB.
- (4) The following parts may be disconnected for the test, either because of their high charging current or because of their effect on voltage limitation :
 - a. Generator
 - b. Power transformer and voltage transformers
 - c. Surge arresters

If the condition of the above parts is in the CGIB, a removable link shall be included in the design of each of the above cases. The design shall be such as to minimize the risk of introduction of faults during the connections after the tests.

The high-voltage tests shall include the following tests :

- (1) AC voltage test
- (2) Lightning impulse voltage test
- (3) Switching impulse voltage test

The test voltage shall be 80% of the voltage specified for the respective routine

tests. The AC voltage test shall be performed for a duration of one (1) minute at the specified level. The lightning and switching impulse tests shall each consist of three (3) impulses of each polarity at the specified level.

6. **QUALITY ASSURANCE /QUALITY CONTROL**

The quality assurance/quality control of the CGIB and other primary electrical equipment shall conform to the requirements included in Part 4, Section 2 Subsection 2.7.

The quality assurance/quality control procedure specified shall include, but not be limited to, the followings:

- (1) All the type tests, routine tests and any specials tests performed on the equipment.
- (2) Special precautions taken at the factory to ensure a clean assembly.
- (3) Field installation procedure to ensure a clean assembly.
- (4) All proposed site tests to verify the quality of the equipment.

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Reports of all the type tests, routine tests and any other special tests performed at the factory.
- (2) Reports of all the site tests, including the tests procedures, any failures and remedial actions taken.
- (3) All other documentation required by applicable codes and standards.

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